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MEViT: Generalization of Deepfake Detection With Meta-Learning EfficientNet Vision Transformer

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ABSTRACT Deepfakes are digitally manipulated videos that appear realistic but are actually fake. With the rapid advances in deep generative models, the accessibility and sophistication of such manipulation technologies are increasing, making it more challenging to detect fake content. Different facial forgery techniques result in complex data distributions, and most existing deepfake detection approaches rely on convolutional neural networks (CNNs) that treat the task as a binary classification problem. While these methods achieve high accuracy on specific datasets, their generalization performance across datasets is often poor due to overfitting to manipulation techniques seen during training. In this study, we propose a model called MEViT, which integrates the EfficientNet Vision Transformer with a meta-learning framework to enhance generalization in deepfake detection. Furthermore, we introduce a pair-discrimination loss to push the feature representations of fake samples away from those of real samples, and a domain adjustment loss to reduce domain shifts across different manipulation methods. The MEViT model is trained on a specific manipulation method in the FaceForensics++ dataset and evaluated on other unseen methods from the same dataset. Additionally, we conduct extensive experiments on multiple deepfake benchmarks, including FaceForensics++ and CelebDF-v2, and compare our method with various state-of-the-art approaches to demonstrate its effectiveness.

INDEX TERMS Deep learning, deepfake detection, GAN, generalization, meta-learning, video forensics.

I. INTRODUCTION

The rise of Deepfake technology driven by advances in artificial intelligence, particularly Generative Adversarial Networks (GANs) [1] has enabled the creation of synthetic videos, images, and audio that appear convincingly real but are in fact fake. While such technology holds potential for creative and commercial applications, it also raises serious ethical and societal concerns, including the spread of misinformation, identity manipulation, and online harassment, and significant trust issues in society [2]. Many problems can be mentioned as creating fake news, spreading misinformation, etc. Deepfake detection has become a crucial area of research to counter these risks. Many existing detection methods treat the task as a binary classification problem and rely heavily on deep learning architectures, such as Convolutional Neural Networks (CNNs), to achieve high accuracy on specific datasets. For example, some approaches exploit visual artifacts [3], while others leverage frequency-based signals [4], [5]. Forgery detection was considered a classification task in many works [6], [7], which concentrated on designing network structures to obtain high accuracy for classification problems. The camera noise [8] and biological signals [9]

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have recently been proposed as basic traces of the forgery detection problem. However, these methods suffer from poor generalization when applied to new or unseen Deepfake generation techniques. This limitation is due to the diversity in forgery artifacts introduced by different manipulation methods, leading to significant distributional shifts between training and testing data. In real-world scenarios, retraining on every newly emerging manipulation type is impractical, making generalization to unseen domains a key challenge.

Domain Generalization (DG) techniques attempt to address this issue by training on multiple source domains to improve performance on unseen target domains [10], [11]. DG refers to the challenge of learning a model from multiple source domains such that it can generalize effectively to unseen target domains without any access to their data during training. This setting is particularly relevant to the Deepfake detection task, where the distribution shift between training and testing data can be substantial due to the rapid evolution of manipulation techniques and diverse video generation pipelines. In the context of face forgery detection, DG is crucial because new forgery techniques may emerge frequently, making it impractical to retrain detection models each time. Many DG approaches aim to bridge the domain gap by minimizing intra-domain variability or maximizing inter-domain feature alignment. Common strategies include meta-learning, feature disentanglement, adversarial learning, and invariant risk minimization, each of which attempts to reduce the reliance on domain-specific cues that may not be present in unseen domains. However, these methods are often hindered by the absence of target domain data during training and still struggle when faced with drastically novel manipulation methods. In this work, we leverage a meta-learning framework to simulate domain shifts during training and equip our model with the ability to adapt to new distributions. We aim to address this by incorporating meta-learning and domain alignment techniques that go beyond standard DG practices.

To address the above limitations, we propose MEViT (Meta-learning EfficientNet Vision Transformer), a novel Deepfake detection model designed to generalize effectively to unseen forgery types without requiring access to manipulated samples during training. Inspired by [12], [10], our model integrates meta-learning with a hybrid EfficientNet Vision Transformer backbone based on research of [7] to design a generalizable model, enabling it to adapt better to distributional changes across domains. The flowchart of our proposal is shown in Fig. 1, and the detailed calculation is described in Algorithm 1. The contributions of our proposal are summarized below.

1) We propose MEViT, a Meta-learning EfficientNet Vision Transformer model, which combines EfficientNet Vision Transformer with meta-learning to enhance generalization for forgery detection.
2) Our method introduces two loss functions. Pair-Discrimination Loss (PDL): Encourages the model to learn discriminative features that push apart real and fake samples at the embedding level, and Domain Adjustment Loss (DAL): Reduces the domain gap among various source domains by aligning their feature representations toward a common embedding center.
3) We validate MEViT using IDFF and CDFF benchmarks derived from the FaceForensics++ and Celeb-DF v2 datasets. The results demonstrate that MEViT consistently outperforms state-of-the-art approaches, particularly in cross-domain settings where generalization is most critical.

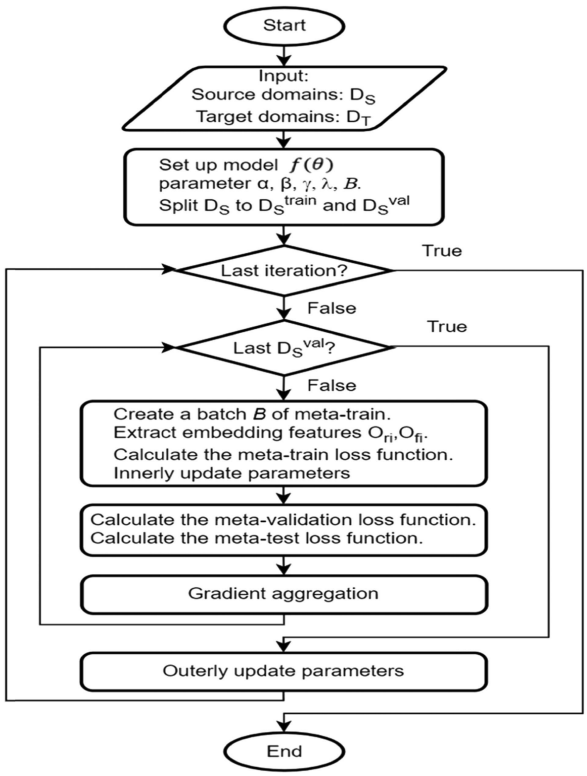


FIGURE 1. The flowchart of the meta-learning-based EfficientNet vision transformer algorithm.

II. RELATED WORKS

In this section, we summarize related work across three main areas: Deepfake creation, Deepfake detection, and meta-learning. First, we highlight key developments in Deepfake generation to provide context for detection challenges. Next, we review significant detection methods, including those aimed at improving generalization to unseen manipulations. Finally, we introduce notable meta-learning approaches, which focus on transferring knowledge across tasks to enable efficient learning in new scenarios.

A. DEEPFAKE CREATION

Presently, there are numerous ways for creating Deepfakes, most of which are variations on GAN [1]. Deepfakes can be created using face-swapping techniques, which have become popular recently. Some relevant studies can be mentioned: MegaFS [13] is a proposed method, which preserves more

facial characteristics and organizes face representation hierarchically in an extended latent space. MobileFaceSwap [14] presented a lightweight network for subject-independent face swapping that dynamically adjusts model parameters based on identity information to swap faces. FSGANv2 [15] proposed a deep learning approach that compensates for considerable posture and expression changes and can be used on pairs of faces without the need for fine-tuning on those faces. For the 3D face swapping method, HifiFace [16] introduced a method that can effectively retain the original face form and create identity vectors to control 3D shape geometric supervision.

Face manipulation is another prominent and extensively useful method for creating Deepfakes nowadays. The face manipulation technique involves training a model on a huge dataset of face motions and expressions and then utilizing that model to synthesize new facial expressions and movements onto an existing video. SemanticStyleGAN [17] proposed a generator that learns to represent local semantic elements individually and then composes pictures. The structure and texture of various local sections are determined by latent coding. StarGANv2 [18] presented an image-to-image translation model that learns a mapping between distinct visual domains while keeping the picture variety and scalability across many domains. BlendGAN [19] is a method for utilizing a versatile mixing method and a general artistic database. InterFaceGAN [20] presented a method to investigate the features of facial semantics embedded in the latent space and recognize the separated face representation obtained by GAN models. Overall, the Deepfake generation technologies are at an advanced stage of development and can produce high-quality face forgeries.

B. DEEPFAKE DETECTION

Deepfake detection involves identifying whether digital media, such as images or videos, has been manipulated using deep learning techniques. Various methods have been proposed, particularly CNN-based models [3], [6] which often achieve high accuracy on known datasets but suffer from overfitting and poor generalization to unseen forgeries. Notable examples include attention-based architectures [21], two-stream models combining semantic and noise features [22], and frequency-based approaches [23], [24]. Other methods leverage spatiotemporal features [25] or person-specific motion patterns [26]. While these models introduce diverse strategies, most rely on closed-set settings and complex architectures prone to overfitting, limiting their real-world generalization.

To obtain good generalization and interpretability, researchers have tried to identify the fundamental difference between the actual and manipulated faces to address the above issues. About studies toward the generalization of Deepfake detection. Some works can be mentioned as PrePGAN [27] proposed frequency-level perturbation maps to make the produced pictures indistinguishable from the genuine ones. This proposal aims to generalize both known and unknown forged images. The study of Luo et al. [28] developed a generalizable model by using high-frequency characteristics and simulating the correlation and interaction among the high-frequency modalities. Forensic Transfer [29] which applied domain adaptation to detect forged faces. They proposed a unique neural network architecture based on an auto-encoder, which is capable of transferring between multiple manipulation domains and learning a model for a derived forensic embedding. The work of Tariq et al. [30] proposed a model named CLRNet. It employs a novel model training technique and investigates both spatial and temporal information in the generalization of detection. Our proposal also focuses on the generalization of Deepfake detection. We apply meta-learning combined with deep convolutional neural networks to handle the issue of generalizing face forgery detection, which is the fundamental aspect that we differ from these works. LGrad [31] which is a detection method for recognizing GAN-generated images. To identify the authenticity of an image, it employs a pre-trained CNN to convert it into gradients that indicate generic artifacts. The gradients are used to depict GAN-generated artifacts by reorienting the attention from data dependency to transformation-model dependency. D3 framework [32] proposed a method that improves the detection of universal artifacts from multiple generators by processing deformed images as discrepancy signals using a parallel network branch. The work of Zhai et al. [33] presented a weakly-supervised method for detecting image tampering that relies solely on binary image-level labels for training. It suggests weakly-supervised self-consistency learning to improve generalization and inter-patch consistency to detect manipulation areas. The hierarchical fine-grained framework for representation learning is proposed in HiFi-Net [34], allows for feature learning and detailed classification by representing forgery features with multi-level labels. The framework includes a multi-branch feature extractor and modules for pixel-level forgery localization and image-level detection. FreqNet [35], a frequency-aware method to improve the generalizability of deepfake detectors. This proposal used convolutional layers on the phase and amplitude spectra between Fast Fourier Transform and Inverse Fast Fourier Transform to capture source-agnostic characteristics.

C. META-LEARNING METHOD

Meta-learning is creating algorithms that can learn from prior learning experiences to adapt to new tasks or challenges more effectively. The purpose of meta-learning is to allow models to learn and acquire information more efficiently and effectively. Many notable studies on meta-learning include: MAML [36] presented a meta-learning algorithm that can be immediately used to model that has been developed using the gradient descent approach. MLDG [10] proposed a training procedure for domain generalization and improving domain shift. The research of Lee et al. [37] allows the meta-learner to learn a subspace on the activation space of each layer where the learner executes gradient descent. The research of Vuorio et al. [38] presented a multimodal MAML method that allows faster adaptation and changes its meta-learned prior based on a task. Recent advances in contrastive learning and model distillation

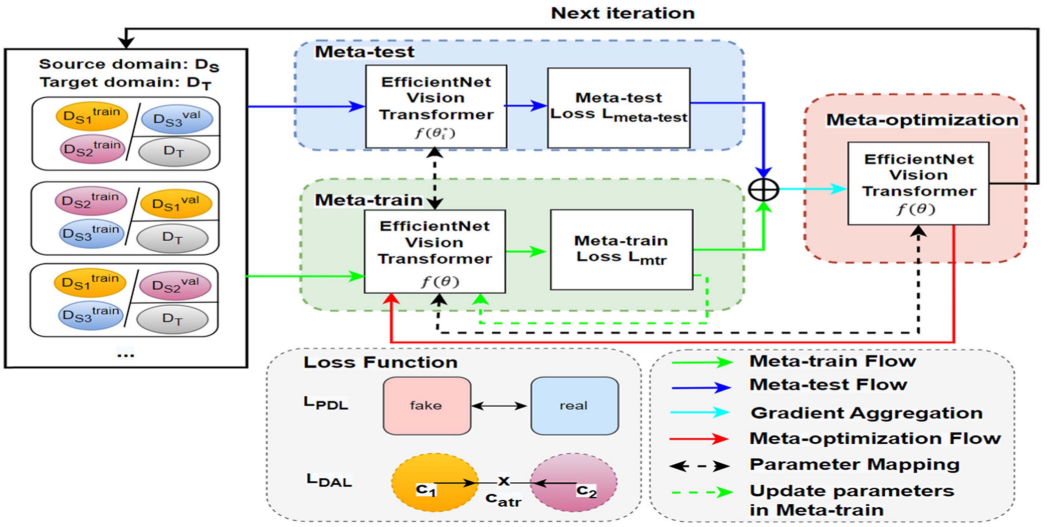


FIGURE 2. The overview of the meta-learning-based EfficientNet vision transformer MEViT.

have also contributed to improving generalization in various domains. For instance, in [39] the authors propose a federated framework combining contrastive learning with feature-level distillation to improve recognition performance across non-IID data distributions.

III. META-LEARNING EFFICIENTNET VISION TRANSFORMER

We propose a Meta-learning EfficientNet Vision Transformer (MEViT) to enhance the generalization of the detector model and the efficiency of identifying manipulated videos. The feature learning component of the MEViT extracts the characteristics of the input images, and the meta-learning method aims to generalize the model across domains. The meta-learning method transforms the model into a meta-learner with strong generalization abilities. It picks up knowledge from several related domains. We consider the EfficientNet Vision Transformer as a backbone $f(\theta)$, the parameter θ needs to be optimized. In the training phase, N domains present data from N datasets. We have $D_S = \{D_{S1}, D_{S2}, \ldots, D_{SN}\}; N > 1$. In the testing phase, the M target domains are used to evaluate unseen domains $D_T = \{D_{T1}, D_{T2}, \ldots, D_{TM}\}; M \geq 1$. Fig. 2 and Algorithm 1 illustrate the proposed MEViT architecture. The source domains and target domains are separated into meta-train, meta-validation, and meta-target domains. A detailed explanation of the separation strategy for these domains is mentioned in Section III-A. The meta-train domains are used to update the model $f(\theta)$ in the meta-train stage, while the meta-validation and the meta-target domains are evaluated in the meta-test stage. Acquired loss functions from both the meta-train and the meta-test stage are applied for the entire updating in the meta-optimization stage. Section III-B illustrates the structure of the model $f(\theta)$. The loss functions that are used in our proposal are introduced in section III-C. The details of meta-train, meta-test, and meta-optimization are explained in Section III-C.

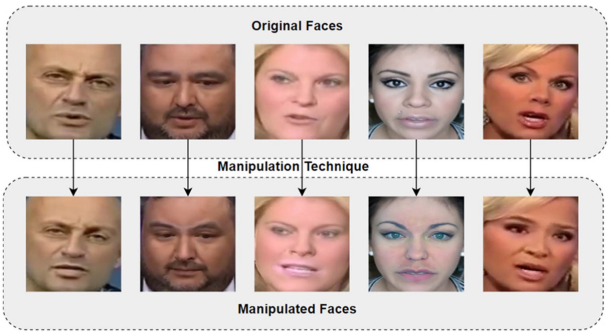


FIGURE 3. Example of original and manipulated faces from FaceForensics++ [40].

A. DATA PREPARATION

Numerous data are used for training the models. Therefore, the ability to learn and make predictions requires extensive dataset preparation. In our work, we build experiment benchmarks on recently available datasets, including FaceForensics++ [40] and Celeb-DF-v2 [41]. Both manipulated and authentic videos, along with their labels (real/fake), are included in these datasets. Since training requires images, these videos are sampled to acquire images. The only part of the image that shows proof of fabrication is the human face, which is mentioned in related studies [42], [43]. Therefore, the MTCNN [44] the library is used to extract faces from images and then resize them to $224 \times 224 \times 3$. A few instances of extracted faces are shown in Fig. 3.

To assess the generalizability of the model, the extracted faces from the above datasets are allocated into source domains and target domains. So in order to accomplish domain generalization, the source domains are randomly divided into the meta-train domains and the meta-validation domains (D_s^{train}, D_s^{val}). The N source domains of D_s^{train} include real and fake face combinations. Specifically, N source domains

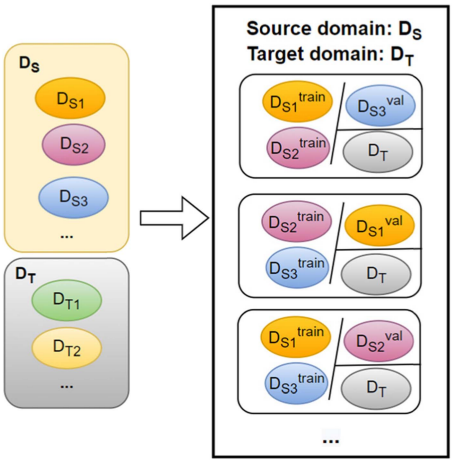


FIGURE 4. An example of a domain separation strategy for the MEViT model. The domains of various datasets are shown on the left, the right part represents the outcome after allocating the source and target domains into different benchmarks.

of D_S are divided into $N-1$ domains for the meta-train and 1 domain for the meta-validation, as illustrated in Fig. 4. The goal of the model is to learn adaptable knowledge and generalize effectively on new domains with various distributions. This setting is used to simulate how domain adaptation works when applied in practical settings. The network can more effectively acquire different features during optimization.

B. FEATURE LEARNING OF EFFICIENTNET VISION TRANSFORMER

The feature learning element follows the structure of the EfficientNet Vision Transformer which is mentioned in the work of Coccomini et al. [45]. The EfficientNet [46] is a convolutional module for feature extraction and Vision Transformer is working as an attention module with a transformer encoder. The EfficientNet specifically generates a visual feature from the incoming face into a chunk. Each chunk is 7×7 pixels. After a linear projection of flattened patches, a Vision Transformer then performs additional processing on each feature from each spatial position. The binary classification value is created using the CLS token. We use EfficientNetB4 as a feature extraction module. The pre-trained weights (IMAGENET) are initialized in the EfficientNetB4 to enable compatibility of EfficientNetB4 output with correlated low-level features of input of Vision Transformer architecture. The entire network will be fine-tuned. This allows the model to fully adapt its pre-trained features to the specific characteristics of the dataset, which can lead to optimal performance, especially when the source and target domains in our setting differ significantly. The Vision Transformer part is a transformer model based on the work of [47]. The training of the Vision Transformer is made simpler by extracting features from the EfficientNetB4 convolutional network because CNN features already include crucial low-level and localized information from the picture. The architecture of the EfficientNet Vision Transformer is shown in Fig. 5.

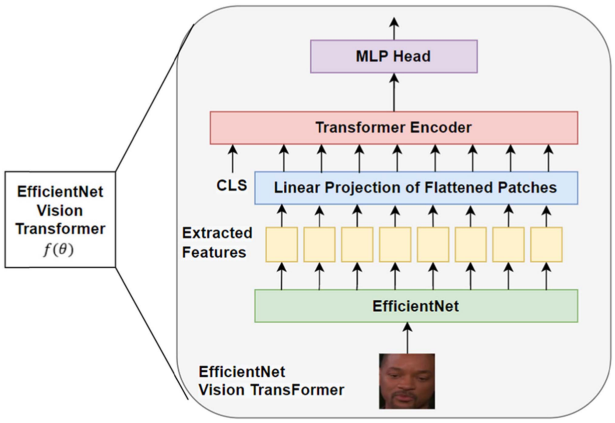


FIGURE 5. The EfficientNet vision transformer architecture.

C. LOSS FUNCTION

We optimize multi-domain distributions so that similar features are mapped into adjacent representations and distinct features are mapped separately from one another to accumulate back-propagated gradients in each batch. We propose two components of the loss function to optimize the model across domains and learn discriminative features.

Pair-Discrimination Loss (PDL): The goal of Pair-Discrimination Loss (PDL) is to push the representation of features of fake samples away from features of real samples while focusing on the optimization of fake and real pairs. A batch of B identities are randomly selected in each iteration and each batch contains the fake and real pairs. In batch B, F is the number of fake samples and R is the number of real ones. This means in each batch $R = \mathrm{F} = B/2$. We declare that the input is X ($X = X_r + X_f$, where X_r is the real input samples and X_f is the fake input samples). An identity B includes $O_r = f(X_r, \theta) = \{O_{r1}, O_{r2}, \ldots, O_{rR}\} \in R^{R\times 2}$ and $O_f = f(X_f, \theta) = \{O_{f1}, O_{f2}, \ldots, O_{fF}\} \in R^{F\times 2}$, O_r is the output of EfficienNet Vision Transformer model $f(\theta)$ with the real faces and O_f is the output of model $f(\theta)$ with the fake faces. The mean embedding feature of the real faces is $\bar{O}_r$, with $\bar{O}_r = \frac{1}{R}\sum_{i=1}^{R} O_{ri} \in R^2$. The Pair-Discrimination Loss (PDL) is formulated as (1). The first term of (1) encourages the embeddings of real and fake pairs to be distinct by maximizing their Euclidean distance. The second term of (1) encourages fake embeddings to deviate from the mean real embedding, ensuring clear discrimination between real and fake clusters.

$$L_{PDL} = -\left[\frac{1}{R}\sum_{i=1}^{R}\|O_{ri} - O_{fi}\|_2 + \frac{1}{F}\sum_{j=1}^{F}\|O_{fj} - \bar{O}_r\|_2\right] \tag{1}$$

Domain Adjustment Loss (DAL): The goal of Domain Adjustment Loss (DAL) is to narrow the domain distance of various domains. The DAL does not function independently but rather works in conjunction with the Pair-Discrimination Loss (PDL) to iteratively refine the alignment of domain-specific embedding features, ensuring that the model learns a more generalizable representation across different domains.

In this process, the position of the mean embedding features of each domain is not fixed, instead, it changes dynamically over iterations. As the PDL modifies the distribution of embedding features by enforcing greater separability between real and manipulated samples, the mean embedding features of each domain, denoted as c_n^t are also affected, causing the domain centers to shift accordingly with each training iteration. Since DAL operates by minimizing the difference between these domain-specific mean embeddings and a global embedding center c_{atr}^{t-1} computed from the previous iteration, the alignment process is inherently dynamic, the embedding features change each iteration leading to a gradual refinement of the domain centers. This continuous adjustment ensures that domain-specific variations are progressively minimized while maintaining a feature space that remains discriminative under the influence of PDL. Over multiple iterations, the combined effect of DAL and PDL results in a stable embedding space where different domains converge toward a common alignment, ultimately enhancing the model's ability to generalize across different manipulation techniques while preserving the distinctions necessary for effective classification. The Domain Adjustment Loss is defined as follows:

$$c_n^t = \frac{1}{2B}\sum_{i=1}^{B}\left(O_{ri}^{D_{Sn}} + O_{fi}^{D_{Sn}}\right) \tag{2}$$

$$c_{atr}^{t-1} = \frac{1}{N}\sum_{n=1}^{N} c_n^{t-1} \tag{3}$$

$$L_{DAL} = \frac{1}{N}\sum_{n=1}^{N} |c_n^t - c_{atr}^{t-1}|_2 \tag{4}$$

where $O_{ri}^{D_{Sn}}$ is the ith output of EfficienNet Vision Transformer model $f(\theta)$ with the original faces in n^{th} training domain and $O_{fi}^{D_{Sn}}$ is the embedding features of manipulated samples, N is the number of source domains, c_n^t is the mean embedding features of nth training domain at iteration t and c_{atr}^{t-1} is the embedding center at iteration $t-1$. For each domain D_{Sn}, we obtain the embedding features of the real samples $O_{ri}^{D_{Sn}}$ and manipulated samples $O_{fi}^{D_{Sn}}$. These embeddings are obtained from the model $f(\theta)$, which processes the input faces and generates corresponding embeddings. The next step is to compute the mean embedding features c_n^t for each domain. This is done by aggregating both real and manipulated embeddings. This step ensures that both types of samples (real and manipulated) contribute equally to the domain-specific embedding representation. Then, the global embedding center for the next iteration c_{atr}^t is computed, which represents the average of all mean embeddings across the N domains. Finally, the DAL encourages the alignment of mean embedding c_n^t of each domain to the global embedding center c_{atr}^{t-1} by minimizing the distance between them. Although it aligns all embeddings (real and manipulated) to a global center, the DAL has the potential to bring real and manipulated features closer together or collapse category distinctions. Therefore, the DAL is designed to complement other loss functions, such as Pair-Discrimination Loss (PDL) and binary cross-entropy loss, by ensuring that the embeddings are not only discriminative but also aligned across domains. This synergy helps the model achieve higher performance in terms of both classification accuracy and generalization. We add a λparameter to make adjusting easier. The impact of the DAL on model performance is analyzed in Table 7 and Fig. 7.



FIGURE 6. Example of some outputs captured from both the original and manipulated videos when using the MEViT model.



FIGURE 7. Impact of different λ and β on IDFF-2-23 benchmark.

D. LEARNING PROCESS

We separate the entire learning process into three main stages: meta-train stage, meta-test stage and meta-optimization stage. The summarization of the learning process is presented in Algorithm 1. The detailed steps are introduced below.

The meta-train stage: The source domains are randomly divided into the meta-train sets and the meta-validation sets (D_s^{train}, D_s^{val}) before looping in iterations. We only use the meta-train sets D_s^{train} for updating model parameters in the meta-train stage. An important point in the meta-train stage

is that the *i*th meta-train set corresponding to the *i*th meta-validation set is not used for training. This setup is repeated for each meta-validation set until the end of the meta-validation set. The B identities are arranged for each meta-train set and the inner learning rate α to control update speed during meta-training. The meta-train loss function is calculated as follows:

$$L_{mtr} = L_{BCE} + L_{PDL} + \lambda L_{DAL} \tag{5}$$

where λ is a parameter to balance DAL and others, and L_{BCE} is the binary cross-entropy loss.

The meta-test stage: We use the meta-validation set and target domain set for evaluation in this stage. In this step, the domain adjustment loss is not used; only binary cross-entropy loss and pair discrimination loss are used. Because these sets of the meta-test stage are not used to update the model parameters in this step. Furthermore, there is often only one evaluation set for each time. Therefore, the domain adjustment loss measurement is meaningless in this situation. However, the other loss functions in the meta-test stages are also utilized for the following steps to improve the generalization of the model across domains. The loss of meta-test is formulated as below:

$$L_{meta-test} = L_{BCE} + L_{PDL} \tag{6}$$

The resulting loss functions are integrated to calculate and update the gradient descent in the gradient aggregation phase after computing the loss of the meta-train and meta-test stages. Through each loop, the values are collected and used to update the entire system in the following step. The gradient descent is presented as:

$$\nabla g_\theta = \beta \nabla_\theta L_{mtr} + (1-\beta)\nabla_\theta L_{mva} \tag{7}$$

where the term ∇g_θ represents the aggregated gradient used to update the model parameters during the meta-learning process. $\nabla_\theta L_{mtr}$ is the gradient of the meta-training loss, which reflects learning of the model on source domains, excluding the current validation domain. $\nabla_\theta L_{mva}$ is the gradient of the meta-validation loss, calculated on the held-out source domain used as a proxy for unseen domains. The coefficient β is a balancing parameter that controls the influence of the meta-training gradient versus the meta-validation gradient.

The meta-optimization stage: After each iteration, we apply stochastic gradient descent to update and optimize the entire parameter θ. The meta-optimization parameter γ is used to controls the gradient aggregation in meta-optimization stage. The best parameter of the source domain during looping through each domain is selected and applied for the next iteration.

Algorithm 1: Meta-Learning-Based EfficientNet Vision Transformer (MEViT).

Input: Source domains:
$D_S = \{D_{S1}, D_{S2}, \ldots, D_{SN}\}; N > 1$
Target domains: $D_T = \{D_{T1}, D_{T2}, \ldots, D_{TM}\}; M \geq 1$.
Init: Set up a EfficientNet Vision Transformer model $f(\theta)$ of the parameter θ with a pre-trained model EfficientNet. The B identities, the parameters: inner learning rate α, gradient aggregation weight β, meta-optimization parameter γ, and parameter for the DAL λ.
Split: $D_S \rightarrow D_s^{train},\ D_s^{val}$
Meta-train: $D_s^{train} = \{D_{S_1}^{train}, D_{S_2}^{train}, \ldots, D_{S_N}^{train}\}$
Meta-validation: $D_s^{val} = \{D_{S_1}^{val}, D_{S_2}^{val}, \ldots, D_{S_N}^{val}\}$
For each iteration **do**
  For each $D_{S_i}^{val}$ in D_s^{val} **do**:
    The meta-train stage:
    Create a batch B identities of meta-train $D_{Sj}^{train},\ j \neq i$,
    Extract embedding features from the model $f(\theta)$: $O_{ri},\ O_{fi}$.
    Calculate the meta-train loss function:
      $L_{mtr} = L_{BCE} + L_{PDL} + \lambda L_{DAL}$
    Update model parameters by:
      $\theta^* = \theta - \alpha \nabla_\theta L_{mtr}$
    The meta-test stage:
      Evaluate with meta-validation D_{Si}^{val}.
      Calculate the meta-validation loss function.
      $L_{mva} = L_{BCE}^{val} + L_{PDL}^{val}$
      Evaluate with D_T.
      Calculate the meta-target loss function.
      $L_{mte} = L_{BCE}^{T} + L_{PDL}^{T}$
    Gradient aggregation:
      $\nabla g_\theta = \beta \nabla_\theta L_{mtr} + (1-\beta)\nabla_\theta L_{mva}$
      $g_\theta^* \leftarrow g_\theta + \nabla g_\theta$
  end
  The meta-optimization stage:
  Update $\theta_{new} \leftarrow \theta_{old} - \gamma g_\theta^*$ by SGD
End

IV. EXPERIMENTS

In this section, we introduce our experiment settings, implementation details, and evaluation benchmarks. To demonstrate the effectiveness of the proposed methods and the generalization ability of the MEViT for unseen domains, we present extensive experiments.

A. DATASETS AND EXPERIMENT SETTINGS

Dataset: Table 1 shows the details of the evaluation benchmarks, which are built based on these datasets. Two public datasets are used in our experiments. Firstly, FaceForensics++ [40]: It includes a total of 1000 real videos and 4000 fake videos produced by four distinct forgery techniques: DeepFake (DF), Face2Face (F2F), FaceSwap (FS), and NeuralTexture (NT). Face-swapping techniques include DeepFake and FaceSwap, while face-reenactment techniques include Face2Face and NeuralTexture. Each technique contains 1000 fake videos, which are separated into three sets: 720 training videos, 140 validating videos, and 140 testing videos. The full dataset is compressed to form two additional datasets (named

TABLE 1. An Example of a Table

Name	Source Domain	Target Domain	Compression
IDFF-1	DF/F2F/FS/NT	DF/F2F/FS/NT	c23
IDFF-2-2 3/40	DF, F2F	FS, NT	c23 and c40
IDFF-3-2 3/40	FS, NT	DF, F2F	c23 and c40
CDFF-1	DF, F2F, CelebDF-v2	FS, NT	c23
CDFF-2	FS, NT, CelebDF-v2	DF, F2F	c23
CDFF-3	FaceForensics++	CelebDF-v2-Merge	c23
CDFF-4	FaceForensics++	CelebDF-v2	c23

Summary of seven benchmarks we build to evaluate our proposal. IDFF: intra-dataset-FaceForensic++, CDFF: cross-dataset FaceForensic++.

c23 and c40) with two different compression levels. The constant rate factor employed in compression is represented by the numbers 23 and 40; a higher constant rate factor indicates lower video quality. Consequently, the proposed approach will be trained and evaluated on two different FaceForensic++ versions (c23 and c40). In each forgery technique of the FaceForensic++ dataset, we use 720 training videos for source domains and 140 testing videos for target domains. In case only one technique is used in the source domain (IDFF-1 benchmark), the training videos of that technique are divided into 3 parts to simulate 3 different domains. Secondly, CelebDF-v2 [41]: The CelebDF-v2 provides a large-scale Deepfake video dataset. It contains 890 original videos and 5639 high-quality manipulated videos divided into 6011 training videos and 518 testing videos. In the CDFF benchmark, we use the training videos for source domains and the testing videos for target domains. Exceptionally, we use CelebDF-v2-Merge for the target domain in the CDFF-3 benchmark. It contains a total of 1500 videos, of which 518 videos are from the test set and 982 videos are selected from the training set of the CelebDF-v2.

Implementation Details: In the initial setup, we pre-train the EfficientNet Vision Transformer with the CelebDF-v2 training set and use that pre-trained model for these benchmarks. The B identity is set as 8. The meta-train learning rate α is set to be 0.0005, the parameter β for controlling the loss function is 0.5, the γ parameter is 0.0005 and the λparameter for controlling DAL is set to be 0.01. As listed in Tables 3, 4, 5 and 6, MLDG [10], Nguyen, et al. [9], Multi-Attentional [42], MAML [36], Learning-to-Weight [11], LGrad [31], HiFi-Net [34] and FreqNet [35] are adapted and re-run with our benchmarks. The pre-trained EfficientNet model (FT-Eff [46]) is fine-tuned on our benchmarks. The other results in Tables 2 and 6 are cited from papers.

Evaluation Metrics: We use the ACC (accuracy score, the higher the better) and AUC (area under the Receiver Operating Characteristic curve, the higher the better) to evaluate generalization capacity, which is widely used in recent forgery detection works.

TABLE 2. Evaluation Results on the IDFF-1 Benchmark

Source domain	Method	Target domain			
		DF	FS	F2F	NT
DF	Face X-ray [48]	0.987	0.633	0.600	0.698
	XceptionNet [40]	0.993	0.736	0.490	0.736
	SFFExtractor [49]	**0.994**	0.500	0.529	0.615
	MEViT	0.991	**0.802**	**0.702**	**0.822**
FS	Face X-ray [48]	0.458	0.961	0.981	0.957
	XceptionNet [40]	0.664	0.888	**0.994**	0.713
	SFFExtractor [49]	0.520	0.985	0.528	0.562
	MEViT	**0.773**	**0.995**	0.923	**0.963**
F2F	Face X-ray [48]	0.630	0.984	0.938	0.945
	XceptionNet [40]	0.803	**0.994**	0.762	0.696
	SFFExtractor [49]	0.736	0.503	0.984	0.562
	MEViT	**0.839**	0.989	**0.989**	**0.961**
NT	Face X-ray [48]	0.705	0.917	**0.910**	0.925
	XceptionNet [40]	0.799	0.813	0.731	**0.991**
	SFFExtractor [49]	0.887	0.506	0.652	0.973
	MEViT	**0.895**	**0.928**	0.819	0.975

The evaluation metric is AUC.

TABLE 3. Evaluation Results on the IDFF-2-23 and IDFF-2-40 Benchmarks

Method	IDFF-2-23		IDFF-2-40	
	AUC	ACC	AUC	ACC
FT-Eff [46]	0.711	0.670	0.617	0.599
MLDG [10]	0.745	0.681	0.613	0.591
Nguyen et al. [9]	0.758	0.705	0.615	0.561
Multi-Attentional [42]	0.742	0.691	0.625	0.596
MAML [36]	0.701	0.668	0.620	0.601
Learning-to-Weight [11]	0.762	0.715	0.651	0.612
MEViT	**0.783**	**0.729**	**0.677**	**0.637**

TABLE 4. Evaluation Results on the IDFF-3-23 and IDFF-3-40 Benchmarks

Method	IDFF-3-23		IDFF-3-40	
	AUC	ACC	AUC	ACC
FT-Eff [46]	0.719	0.651	0.623	0.556
MLDG [10]	0.760	0.698	0.675	0.611
Nguyen et al. [9]	0.742	0.667	0.672	0.606
Multi-Attentional [42]	0.748	0.687	0.669	0.594
MAML [36]	0.691	0.657	0.642	0.602
Learning-to-Weight [11]	0.772	0.705	**0.689**	**0.644**
MEViT	**0.802**	**0.727**	0.682	0.627

B EXPERIMENTAL RESULTS

To properly assess the generalization capacity of the proposed model, we design extensive cross-database evaluations including seven benchmarks. Table 1 shows detailed information on each benchmark. There are two kinds of cross-validation settings, i.e., intra-dataset FaceForensic++ validations (IDFF), which train and test with different techniques in the FaceForensic++ dataset. Secondly, cross-dataset Faceforensic++ validation (CDFF), which evaluates with both the FaceForensic++ and CelebDF-v2 datasets.

As illustrated in Table 2, we apply manipulated images of one technique for the source domain and those of all four techniques for testing with the target domain, respectively. In most cases, our model achieves better results than other approaches, especially for the unseen techniques (the techniques are in the

TABLE 5. Evaluation Results on the CDFF-1, CDFF-2 and CDFF-3 Benchmarks

	Method	AUC	ACC
CDFF-1	FT-Eff [46]	0.731	0.659
	MLDG [10]	0.765	0.724
	Nguyen et al. [9]	0.721	0.679
	Multi-Attentional [42]	0.745	0.689
	MAML [36]	0.725	0.704
	Learning-to-Weight [11]	0.769	0.722
	MEViT	**0.783**	**0.735**
CDFF-2	FT-Eff [46]	0.717	0.633
	MLDG [10]	0.762	0.678
	Nguyen et al. [9]	0.725	0.645
	Multi-Attentional [42]	0.759	0.666
	MAML [36]	0.689	0.625
	Learning-to-Weight [11]	0.782	0.711
	MEViT	**0.798**	**0.721**
CDFF-3	FT-Eff [46]	0.605	0.566
	MLDG [10]	0.658	0.601
	Nguyen et al. [9]	0.601	0.551
	Multi-Attentional [42]	0.615	0.585
	MAML [36]	0.631	0.596
	Learning-to-Weight [11]	0.681	0.623
	LGrad [31]	0.622	-
	HiFi-Net [34]	0.679	-
	FreqNet [35]	0.685	-
	MEViT	**0.705**	**0.648**

TABLE 6. Evaluation Results on the CDFF-4 Benchmark and Comparisons With Recent Works

Methods	Training Dataset	AUC
XceptionNet (c40) [41]	FaceForensics++ (c40)	0.655
XceptionNet (c23) [41]	FaceForensics++ (c23)	0.653
LRNet [50]	FaceForensics++ (raw)	0.574
Nirkin et al. [51]	FaceForensics++ (c40 and c23)	0.660
CNN+RNN [50]	FaceForensics++ (raw)	0.615
Schwarcz et al. [52]	FaceForensics++ (c23)	0.633
MTD-Net [53]	FaceForensics++ (c23)	0.701
Multi-Attentional [42]	FaceForensics++ (c23)	0.674
LGrad [31]	FaceForensics++ (c23)	0.665
HiFi-Net [34]	FaceForensics++ (c23)	0.742
FreqNet [35]	FaceForensics++ (c23)	0.723
MEViT	FaceForensics++ (c23)	**0.753**

target domain but not in the source domain). This is because the Pair-Discrimination Loss enhances the model's ability to distinguish real and fake samples, while EfficientNet-ViT captures both local and global visual features. These advantages enable MEViT to outperform CNN-based methods, which often overfit to specific manipulation patterns. Face X-ray [48] has a considerably good generalization ability since it identifies blending evidence. XceptionNet [40] excessively depends on texture patterns, its performance suffers greatly in unknown forgeries. The approach of SFFExtractors [49] obtains quite good outcomes as well. In some cases, the higher AUC is noted by SFFExtractors [49], they adopt a different solution.

In the benchmark IDFF-2 and IDFF-3, the experiments are separately performed on high-quality and low-quality videos. The settings in these benchmarks are to evaluate generalization from different techniques in FaceForensic++ dataset. As shown in Tables 3 and 4, in most cases, our method achieves better results compared to other methods. Its success lies in its **meta-learning framework**, which simulates domain shifts during training, allowing the model to learn more generalizable features. This leads to better cross-manipulation robustness than methods relying solely on supervised learning. The FT-Eff [46] model has been fine-tuned using our benchmarks, which detect forged images but do not generalize well for the target domains. The MLDG [10] and MAML [36] proposed a basic meta-learning approach that should be generalizable but not specialized for Deepfake detection, resulting in a slightly lower result. Different methods are used in the Multi-Attentional [42], Learning-to-Weight [11] and the research of Nguyen et al. [9]. Each method proposes a unique solution for detecting Deepfakes and generalizing the model. In the IDFF-3-40 benchmark, the performance of the MEViT model is slightly lower than Learning-to-Weight [11]. This demonstrates that the MEViT model is rather sensitive and suffers from lower-quality videos.

The settings of CDFFs are more challenging than IDFFs. Table 5 shows that our proposal results in notable improvements. The results show that our approach enhanced the generalization of the model. These results are driven by its ability to learn domain-invariant features, which allow it to generalize across datasets. Traditional methods lack this capability, making MEViT especially effective for cross-dataset Deepfake detection. Learning-to-Weight [11] is a strong competitor, performing close to MEViT in some cases but falling short overall. This method is designed to detect fake faces and then use a meta-learning technique that simultaneously optimizes the weight-aware network and uses meta-optimization to handle source/target domain shifts. This leads to quite promising results when cross-evaluating with numerous datasets. Traditional meta-learning methods like MAML [36] and MLDG [10] lag behind, indicating the need for more advanced approaches in these challenging datasets. The lower classification performance of these methods can indeed be attributed to the fact that these techniques are not specifically designed for face forgery detection problems. Face forgery detection involves unique challenges, such as subtle manipulations in facial textures, expressions, and fine-grained image features, which demand specialized approaches tailored to this domain. Performance decreases in CDFF-3 for all methods suggest that it presents the most significant challenges. Even with the examination of new techniques like LGrad [31], HiFi-Net [34], and FreqNet [35] for assessing this difficult standard, the outcomes are still unsatisfactory.

To make a fair comparison, we further compare our proposal to state-of-the-art works that use similar settings in forgery detection. As shown in Table 6, we compare our strategy to XceptionNet [41], a two-stream Recurrent Neural Network – LRNet [50], the research of Nirkin et al. [51], CNN+RNN [50], the research of Schwarcz et al. [52], MTD-Net [53], Multi-Attentional [42], LGrad [31], HiFi-Net [34] and FreqNet [35]. The methods XceptionNet (c40) [41], XceptionNet (c23) [41], LRNet [50], Nirkin et al. [51], CNN+RNN [50], and Schwarcz et al. [52] were developed

based on traditional CNN classification models. While these methods often achieve strong results in intra-dataset evaluations, their performance tends to decline when applied to generalize across multiple datasets. MTD-Net [53] used the multi-scale texture difference of fake samples approach, but encountered challenges when training the model on data with a broader distribution of fake categories. Multi-Attentional [38] employs attention algorithms to focus on fake attributes. However, the results reveal that these fake attributes are sometimes misidentified when applied to new data domains. LGrad [31] produced suboptimal results when evaluated using the cross-evaluation CDFF-4 benchmark. This indicates that LGrad performs effectively in intra-dataset evaluations. As demonstrated by the authors, the method is designed to handle a diverse range of categories rather than focusing exclusively on faces. This broader scope may lead to reduced performance when evaluating faces under more challenging constraints, HiFi-Net [34] achieves remarkable results, supposedly due to its architecture, which incorporates a multi-branch feature extractor and performs fine-grained classifications across different attribute levels. These classifications also assist in localizing forgeries. FreqNet [35] approaches the problem through the frequency domain, and while the results show that this approach shows promise for generalizing across various data domains, it is not recommended for domain adaptation problems. This is likely due to the loss of certain critical connections between data domains during the conversion to the frequency domain. The results show that our proposal outperforms the existing works. The advantage stems from the synergy of transformer-based representation learning, meta-learning optimization, and custom loss functions (PDL + DAL). These components collectively equip MEViT with strong generalization even on unfamiliar datasets and manipulation methods.

The outcomes when running the MEViT model on videos from the dataset are shown in Fig. 6. The results of the real and fake images are displayed on the top and bottom, respectively. The manipulation techniques are so subtle that it is difficult to identify them with normal eyes.

C. ABLATION STUDY

The effect of λ and β on our model is illustrated in Fig. 7. The λparameter is to control the DAL with other losses, this parameter adjusts the correspondence of different domains. The β parameter is to balance training loss and testing, validation losses. We evaluate our method on the IDFF-2-23 with AUC and ACC metrics. The best result of λ is 0.01 while the best result of β is 0.6.

To evaluate the impact of different loss components, we conduct an ablation study of our model on IDFF-2-23. Table 7 shows the efficiency of each performance component. When only using binary cross-entropy loss, the acquired results are reduced to the lowest. Better results are obtained when applying PDL with our model and DAL with our model. This demonstrates the representation adjusting of fake and real sample features of PDL and the ability to align different domains of DAL. The model reaches the best performance when applying both DAL and PDL, showing their complementary effect in improving cross-domain robustness and detection accuracy. By aligning the embeddings of real and manipulated samples to a common feature space, DAL enhances the ability of the model to generalize to unseen manipulation techniques. This is crucial in domains where new, previously unseen manipulation methods can emerge. DAL helps the model become more robust to variations in the feature distribution across manipulated samples. The key advantage of DAL is its ability to find domain-invariant features. The loss function ensures that the model learns features that are not tied to any specific domain, making the learned representations more universally applicable.

TABLE 7. Ablation Study of Our Proposal on the IDFF-2-23 Benchmark With Pair-Discrimination Loss (PDL) and Domain Adjustment Loss (DAL)

PDL	DAL	AUC	ACC
-	-	0.723	0.633
✓	-	0.749	0.689
-	✓	0.754	0.701
✓	✓	**0.783**	**0.729**

TABLE 8. Performance Comparison of Our Proposal on the IDFF-2-23 Benchmark with Different Backbones

Backbone	AUC	ACC	# params
MEViT-B0 (EfficientNetB0)	0.763	0.694	109M
MEViT-B4 (EfficientNetB4)	0.783	**0.729**	122.7M
MEViT-B7 (EfficientNetB7)	**0.801**	0.725	169.7M

Table 8 presents a performance comparison of the proposed method using three EfficientNet backbones on the IDFF-2-23 benchmark. In this experiment, we use a basic configuration utilizing all loss functions. The results show that EfficientNetB7 achieves the highest AUC, which indicates that the model with the EfficientNetB7 backbone has the best overall ability to distinguish between classes. EfficientNetB4 achieves the highest ACC, meaning the model with this backbone has the best accuracy in classifying the target samples correctly. EfficientNetB0 has the lowest performance, showing that the smaller backbone provides less discriminative power. There is a trade-off between AUC and ACC. While EfficientNetB7 has the highest AUC, its accuracy is slightly lower than that of EfficientNetB4. The choice of backbone should be based on the specific task requirements. However, EfficientNetB7 requires significantly more training time and higher hardware resources compared to EfficientNetB4. Due to the extensive experiments in this study, EfficientNetB4 was chosen for the majority of the experiments.

Computational Complexity Analysis: We evaluate the computational cost of MEViT using three EfficientNet backbones (B0, B4, and B7) to assess the trade-off between performance

and efficiency. Table 8 summarizes the number of parameters for each backbone configuration. MEViT-B4 offers the best trade-off between performance and computational cost, making it suitable for real-world applications with limited resources. Despite incorporating a Vision Transformer module, MEViT remains computationally efficient due to the lightweight yet expressive EfficientNet backbone and the use of compact image patches.

V. CONCLUSION AND FUTURE WORK

In this article, we proposed a Meta-learning EfficientNet Vision Transformer model to enhance the generalization of face video forgery detection methods. We also introduced the pair-discrimination loss is to move features of fake samples away from features of real ones while focusing on optimization of fake and real pairings and the domain adjustment loss is to reduce the domain distance across different domains. When integrated with the learning process, the proposed model improves domain adaptation and generalizes well with new domains. The extensive experiments indicate the efficiency of each module, and the comparisons with related methods verify the generalization capacity of our proposal.

Developmental Trends and Challenges in Deepfake Detection. Deepfake detection research is undergoing a notable shift from conventional CNN-based supervised classifiers toward more generalizable and robust learning paradigms. There are some limitations and potential risks to the proposal. Particularly, the method relies solely on visual information. However, it is increasingly common to manipulate or synthesize audio content to produce false or misleading audio recordings. The original audio may still be present in Deepfake videos that have just undergone visual editing, providing the audio the opportunity to provide listeners with false contextual cues. Deepfake detection using only visual information becomes more and more difficult as Deepfake creation techniques advance. Emerging trends include the use of transformer architectures, meta-learning, and contrastive or self-supervised learning to improve feature representation across diverse manipulation techniques. There is also growing interest in domain generalization, federated learning, and lightweight models for real-world deployment. However, significant challenges remain: generalization to unseen forgeries is difficult due to rapidly evolving manipulation methods and a lack of labeled data. Additionally, interpretability and explainability of detection results are still limited, making adoption in critical applications more challenging. In future work, we will explore the synchronization of audio-visual in Deepfake detection and detecting inconsistencies between speech and lip movements. Furthermore, we aim to apply multimodal approaches that offer promising avenues for Deepfake detection.

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